

Minimizing change aversion through mixed methods research: a case study of redesigning Spotify's Your Library

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ABSTRACT

Launching a radical change to a habitual feature, used by millions every day, presents a challenge both to the end user and the organization making the change. Change aversion to habitually used features is a known factor in users experience design, but at times major changes are needed to encompass feature growth ultimately benefiting users.

In this case study, we present examples of how data science and user research collaborated during the redesign the mobile Library feature at Spotify. Major adaptations were needed in order to enable future opportunities. The challenge required a high degree of sensitivity to users' needs within what is often considered their space in the world of streaming. We believe that close collaboration between the product, design, engineering, data science, and user research disciplines, with a focus on quantitative and qualitative mixed methods insight work in early development phases, enabled the experience to be launched with the users' experience at the forefront and thus minimising change aversion.

CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**.

KEYWORDS

user experience, mixed methods research, user research, data science, change aversion, audio streaming, library organization

ACM Reference Format:

Ingrid Pettersson, Carl Fredriksson, Raha Dadgar, John Richardson, Lisa Shields, and Duncan McKenzie. 2023. Minimizing change aversion through mixed methods research: a case study of redesigning Spotify's Your Library. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3573875>

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CHI EA '23, April 23–28, 2023, Hamburg, Germany
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ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3573875>

1 INTRODUCTION

In 2008, Spotify changed the way the world experiences music by introducing legal streaming, as a response to the piracy in the music industry. Since that breakthrough, there have been continuous learnings about how the design of the streaming user interface influences the experience and listening habits. Habits are powerful behavior drivers; users are typically reluctant to embrace product changes that will disrupt them in their daily routines, including routines on digital platforms. Disrupting habits can even be linked to perceived losses, and thus cause loss-aversion effects. These perceived losses and disadvantages are even found to have greater impact on users' preferences than gains and advantages to be had [23]. Thus, as redesigning a habitual and loved feature runs the risk of resulting in loss-aversion sentiment, such launches need to be approached with care [20].

Spotify states that *'Our mission is to unlock the potential of human creativity—by giving a million creative artists the opportunity to live off their art and billions of fans the opportunity to enjoy and be inspired by it.'* [21] 'Your Library' is a focal area in Spotify where users return to their liked or followed content, artists and creators. In an app with an abundance of content, the act of curation, saving content to my library, building playlists etc, are strong habits for many users on the platform. We are conscious that changes in a streaming service's library structure thus can have a substantial impact on users' experience [2][7][14]. Approaching redesign of a feature that millions of users use regularly is a challenge given the risk of disrupting users and causing negative sentiment, but may be a necessity for growth and additional user value.

In 2021 Spotify launched a new version of the Library and in this paper we will showcase examples of how we approached the task through insights work. Our goal for this project was to deliver on a framework that reimaged the experience of retrieval and organization for a potentially limitless number of audio formats. In 2019, a previous version of the Library had been launched, where the goal of the redesign was to accommodate Spotify's rapidly growing podcast proposition, inside of a product that up until then had been centered almost exclusively around music. These efforts were successful in elevating the status of podcasts, however it became clear that the 2019 design would not scale as Spotify aimed to transition further into the audio network of the world. In this case study, we will present some of the approaches the product insights team employed in order to reach the ambitious goal of

creating a library for many audio formats. Spotify has long valued the application of multiple complementary research methods across qualitative user research and data science [1], as it is good practice for to provide complementary perspectives on users' experiences, that mutually enhance each other [11]. For the product insights team, the project represented a deepened collaboration with also other competencies, as the data scientist and user researcher were further embedded in the tech and design project than what had previously been the norm.

This paper provides examples of how qualitative and quantitative methods were planned and conducted together in practice to approach a potentially disruptive launch for users, and what learnings we have made from the partnership between qualitative user research and data science. In contrast to the parallel quantitative and qualitative research processes often showcased [18], we present a case where mixed method's insights work was integrated from the get-go. By employing early experiments at scale, we could assess that we were not hurting guardrail metrics such as listening time and user retention. By synchronous qualitative research on the experiences, we could identify where to adapt and refine the experience, providing clear recommendations and status reporting to the team.

In the discussion we reflect on learnings made when working towards our goal to rebuild a Library feature for the future of streaming.

2 BACKGROUND

In order to understand where our largest challenges, risks and opportunities for the Library resided, we needed high quality insights into the current user experience. By combining data science and user research discipline work throughout the process, we sought to gain a holistic understanding of the user experience and mitigate blind spots where possible. By this approach, we strove to increase the directionality of the insights, as decision-making could be achieved with an in-depth understanding of the experiences at hand. Employing quantitative methods together with qualitative methods can offer an understanding of both the size and the underlying reasons of a phenomenon. Furthermore, using mixed methods helps counteract inherent biases from data sources, researchers and methods [5][11]. However, although recognized as a goal to strive for, bridging the gap between quantitative and qualitative crafts in practice is a well-recognized challenge for many researchers, both in industry and academia [11] [18].

In the coming sections we provide examples of how we conducted the work across data science and user research. We also present some of the collaborative aspects in our work, where we worked closely across design, product and engineering teams. For insights to have impact, not only high quality data is needed, but also trust. Passi and Jackson [16] highlight how trust and impact of data science does not only rely on the data and insights' quality alone, but also on the negotiation and translation of the data by the scientists. Bearing this in mind, the data scientists and user researchers were further embedded into the engineering and design organization than previously.

In the early phase of the design process, we sought insights that could inspire and explore new directions as well as frame the



Figure 1: Photo from the field study

underlying mental models of users [24]. These insights fed into strategy development and early design directions. Once concepts were under development, insights sought to rapidly inform and optimize designs in order to reduce risk and improve usability. When designing for interactive experiences, there is also a novelty effect to be accounted for, which can be described as a person's subjective first responses to using a technology [22]; these first responses often do not correspond to the patterns of usage that will persist over time, as the product ceases to be new to the user. Thus, we decided to enroll interview participants into the new user experience proposition and allow them to use it for an extended period of time before interviewing, which enabled us to address experiences in-depth beyond the novelty effects. At late stages of the process, insights aimed to measure product performance and experimentation was in focus to further reduce potential risks and identify any remaining experience gaps before launching the new experience.

3 EXAMPLES OF INSIGHT WORK

3.1 The Meaning of the Library

During the early phases of the development process, product insights can provide an important foundational understanding of user context and needs to frame the project [17]. This was also the case in our work - at the start of the project, user research and data science aimed to understand current key user behaviors, needs, and motivations, in order to inform design and product direction. From data explorations in this phase, we observed key consumption patterns centered around the feature. Specifically, we were also interested in establishing an understanding of the saving and retrieval patterns for different audio format types on the platform, forming a baseline understanding to compare upcoming changes to. While data explorations were made, we conducted a qualitative field study and a semi-structured interview study, both targeting the users' experience and ascribed meaning of the Library. In this way, we could employ explanatory mixed methods research to understand why users were habitually returning to the library. This research was done in order to inspire new directions and opportunities for the Library and to understand the boundaries of change and adoption.

3.1.1 Field Study. In the field study, 18 US participants were interviewed and observed in-home in Atlanta and Philadelphia. The participants were asked to walk the researchers through their Library content and we also asked and observed how they used Spotify in their everyday life and how they compared it to other digital

platforms they used. After each session, we held team debriefs, following ethnographic practices [12]. The user researchers conducted the research sessions with both product managers and designers present, in order to gain mutual understandings of insights and directions, as well as build empathy with users.

3.1.2 Remote User Interviews. Yet another 11 UK participants were interviewed remotely in video calls in order to further drill down on the motivations for personal collections in the age of recommendation systems. Both studies reinforced our assumption that the Library is a highly personal space, where users wish their ownership and integrity to be respected. The artifacts created and collected in their libraries were often charged with emotions, such as break-up playlists, or habitually used, such as the Liked Songs collection. The following user quotes clearly represent the ownership the participants felt for the Library content:

"[My music] I want to feel I have it. I want to feel I know it"

"My Library represents me very much - better than my wardrobe I would say"

3.1.3 User Needs. Precisely as wardrobe use, we also observed very different attitudes and behaviors in relation to organization: some users were keeping their collections in a neat and strict order, while others were gathering large amounts of content to later rely on that one can search and find when needed. However, the foundational mental model of the Library as "their space" in a world of recommendations was ever present. The playlists they create and store can hold much pride, identity and nostalgic value to users, and at times also represent important co-experiences with others, such as collaborative playlists.

Some of the core user needs observed were *Ownership*: wanting control over what is in the Library, *Organization*: ability to address disorganization in the Library, *Retrieval*: having clarity of content location and ease of access.

To summarize, in this first insights phase, we performed the mixed methods groundwork of exploring current behavioral data and the user needs through qualitative methods.

3.2 Approaching Change

"I'd be like, get out of here! That's MY library..."

Being mindful of the issues with disrupting users' core routines in a loved space, we wanted to prepare thoroughly for the redesigned Library before experiments on the platform itself. We conducted several rounds of usability testing of multiple interaction design concepts using interactive prototypes. The interaction design concepts were updated after each iteration. In this way, the researchers enabled key concept decisions to be taken early based on insights. This built confidence in new UX elements introduced. Concept iterations included, for example, new ways to navigate the Library using filters, the ability to search as well as explore different views of the Library content. The designer, data scientist and product manager observed the usability testing sessions and participated in debriefs after the sessions. In this phase, data science and user research collaborated by for example understanding saving and retrieval behaviors of participants, making sure we had representative samples in our studies.

In addition to answering whether we were providing a way for our users to reach their goals effectively and efficiently whilst enjoying the experience [15], the usability testing allowed us to observe patterns in how users were organizing and searching for content in their libraries. Precisely as in the previous interviews, we observed how the Library is experienced as a highly personal and controlled space; users reacted negatively to major intrusion into their Library without their specific permission. Furthermore, participants continued to crave easy retrieval and organization of what they have collected. Some of our concepts scaled less well across users with large collections, and we aimed for a concept that could perform well for both large and limited collections.

3.3 First Exploration of the New Holistic Experience at Scale

To learn about emerging behaviors and user responses to the substantial change of the Library experience, before exposing the redesign to a large group of users, we gave a group of users on Spotify's official community forum the option to try the our new, and substantially different, beta version of Your Library in their Spotify app. Knowing that the prolonged usage of a product can be significantly different than the initial experiences [10], we kept the users in the new experience with enough time for new habits to emerge. From online feedback and interviews from these highly engaged users with typically large collections, we got thorough feedback, both positive and negative. This helped us identify points of friction and to make some changes before letting more users try it.

While gathering feedback from the selected users from the community, the team was building features that would enable a small sample of the user base to opt-in to the beta test. Motivating and explaining a major re-launch of a habitually used feature is important, as Sedley and Müller points out in a previous CHI case study on launching disruptive changes [20]. Hence, we provided onboarding to the beta experience through in-app messaging. When onboarded, the users had the option to give feedback in the form of a rating of how satisfied they were with the experience and an optional text response. As behavioral data was also collected, we were using concurrent method triangulation to assess the user experience. One fifth of the invited users opted-in and while an overwhelming majority of users stayed within the experience, a minority of them sent feedback, with a smaller portion also including text with their submission. Even with the large loss of users in every step of the funnel, we received text feedback in the order of thousands and rating feedback in the order of tens of thousands.

For the analysis of the text feedback, we used machine learning in the form of a LDA topic model [4] to extract topics from feedback texts, and 11 distinct topics were found. We did this in combination with manual analysis of 400 free text feedback chosen randomly, to ensure to capture nuances that may have been missed due to limitations of the themes provided by the LDA topic model. Such nuances were for example the levels of frustration in regards to different aspects of the experience. Combining the two approaches of analysis, as well as data explorations of the users' collection and retrieval habits, proved to give a more accurate and fast understanding of users' experiences which lead to interesting findings.

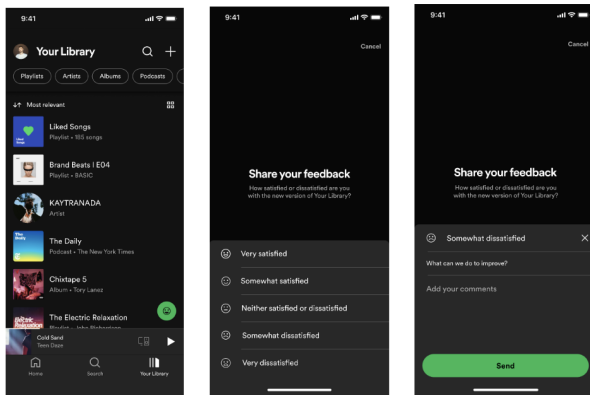


Figure 2: User feedback was enabled in the beta Your Library interface

For example, we learned that although a smaller number of users were frustrated with changes to sorting, they were very passionate and vocal about this change, leading us to reconsider the sorting options.

3.4 Exploring Tweaks to the New Experience

The beta test enabled us to establish an overview of how users perceived the new Library experience at scale, over time. To address improvements, we set up a continued flow of cooperation across data science, user research, design and engineering, with weekly meetings, regular status updates and mixed method insights artifacts, which summarized quant and qual findings.

We categorized pain points from the beta test in three different groups, based on our understanding of the problem and risk assessment. The first group were low-level pain points; well-known issues that required low risk changes to the designs. These design changes were done immediately to address the feedback. The second group of mid-level pain points required changes with more substantial levels of risk and more uncertainty. For these, we had a plan for how to improve the experience, but the changes called for more attention and effort from the team, as the issues contained added complexity. The third group contained high-level pain points. Although identified, these high-level pain points were not possible to address swiftly, as the team still knew little about the motivations and details behind their occurrence. This was expected as the open text feedback from users, by nature, was limiting for in-depth understanding.

The feedback provided direction on what changes to make. However, the beta test only provided observational data, notably from self-selected users, which could not be used to reliably answer the key causal question of “*how will our key metrics be affected if we launch the new experience to the majority of our users?*” Answering this question was of utmost importance before the new experience could be launched, and thus the next step was to start an A/B testing process.

3.4.1 A/B Tests. To address the second group of mid-level pain points, we designed an A/B testing plan where the goal was to

guide the team on additional changes to make before launch and ultimately to evaluate whether the experience was ready for a global launch. Key metrics would be evaluated in the planned tests. These key metrics were divided into two groups: guardrail metrics, that we did not aim to improve, but had to be kept within desired margins before launching, and success metrics that we hoped to improve with the new design. The guardrail metrics focused on listening time and retention for the mobile application. The success metrics were more local and focused on content retrieval from the Library. Note that because of the business value of releasing a Library that enabled additional audio types, the success metrics did not need to show significant improvement before launch. The success metrics were included to provide guidance on what changes to make to the new experience pre-launch and how to develop it further post-launch.

The testing plan started out with a small-scale A/B test containing three groups: a control group with the old Library, a treatment group with the new Library, and a treatment group with the new Library with an additional new feature. The goal of this test was to provide guidance on whether the new feature, aimed to address usability issues, should be included in the new Library design or not, and to estimate the impact of the new Library on guardrail metrics. If the test analysis did not indicate signs of negative impact to the guardrail metrics, the plan was to move on to a larger A/B test with more statistical power. If we observed signs of negative impact, the plan was to pivot and run another small test after additional changes were made.

The small-scale test was successful and the treatment with the new feature added performed better than the treatment without the new feature. Thus we moved on to a larger A/B test containing two groups: a control group with the old Library and a treatment group with the new Library with the new feature added. Since the new Library enabled additional audio types, we were ready to accept some potential negative impact to the guardrail metrics in the short term. However, the potential impact had to be kept within acceptable limits. To manage this, we utilized non-inferiority testing, which is a type of statistical test where the aim is to show that a new product is not unacceptably worse than an older one[19]. Non-inferiority margins (NIMs) are used to formalize what unacceptably worse is. The new Library would be considered non-inferior if we could say with a predetermined level of certainty that launching the new experience to all users would not negatively impact the guardrail metrics beyond these margins. For non-inferiority tests to be valuable, one has to carefully select NIMs that are relevant to making a decision, in this case whether to launch the new Library or not. The main reason for running a larger A/B test before launch was to increase the sample size enough to statistically power the desired NIMs. The larger test was also successful and we started to prepare for a global launch, with the learnings from the following qualitative study in consideration for continued development.

3.4.2 Diary and interview studies. In combination with the A/B tests, we conducted a diary and interview study, addressing the third group of pain points identified (high-level risks). With this qualitative research enabled by the quantitative test, we aimed to understand the experience effects of changing habits over time, as well as the effects of the high-risk pain points. We utilized the A/B

tests to enroll seven diary study participants into the experience, whereby the participants were given access to the new library for one week, with regular check-ins every other day and a final longer interview at the end of the week. This gave them the opportunity to see their own content in the new experience and ensured we didn't react to a person's subjective first responses to using a technology. During the diary study, participants were encouraged to explore their new Library and continue with their habitual listening. The interviews from this diary study gave valuable in-depth understanding of reasons behind the pain points we had previously detected in the beta testing phase, which allowed us to go back to the drawing board with more confidence and re-think aspects of the experience. Examples of redesigns were introducing new podcast collection formats as well as redesigning the retrieval experience and sorting options in the Library.

Additionally, the diary study and interviews revealed interesting findings regarding the core user needs identified in the early research, specifically organization. While organization in the library proved to be a central user need, it became evident that users don't want to spend time and effort organizing their Library, and would rather enjoy listening and discovering content instead. This learning was significant as it challenged a core product hypothesis for the future of the Library and led to the planning of additional qualitative studies after launch to further investigate and form a product strategy more aligned with user behavior.

3.4.3 Launching. Once we were certain through qualitative and quantitative research that the feature would not disrupt key habits, we launched the experience to all users to avoid scattered experiences across the user base. We found that we did not critically disrupt users - they were able to find and consume their collections, as well as utilize the new features added for quicker retrieval. Negative sentiment did exist for the new update, as is expected for changes, but it was substantially lower compared to the previous Your Library update.

4 DISCUSSION

We succeeded in keeping important habits intact around saving and retrieval of content intact, while preparing the Library to adapt to future development. In the following sections we will expand deeper on the lessons learned and limitations to our work, specifically in relation to the qualitative and quantitative research conducted.

4.1 User Needs and Challenging Assumptions

Early on, we observed the key user need of still wanting to be able to collect and curate in the age of recommended music, as well as the desire to experience strong ownership of this collection. These needs are also confirmed by academic research of library experiences on digital platforms, where for example Lüders' [13] user interviews data from the previous Spotify Your Library launch suggested that users value opportunities to explore music, yet their practices are equally directed towards archiving music. As Bhaskar [3] points out, user curation still holds a place on online platforms as it helps to reduce cognitive burden, save time and to overcome 'choice overload'. Precisely as Young (2008) describes, we found that the mental models designed for are more successful when they

align with the users' own mental models, for example users' mental models of expected sort orders for the Library content.

Our journey also included walking back on assumptions around some user needs and how to cater for them. One such instance was that we observed how users typically were overestimating how much effort they would spend on organizational tasks within their Library. Continuously forming hypotheses from early assumptions and testing these hypotheses through quantitative research proved to be a vital step for building a strategy for the future of the product with confidence.

4.2 Collaboration

Our mixed methods approach enabled us to understand if we were underperforming in terms of metrics and user experience. Employing both quantitative and qualitative research on real-life use, over time, enabled us to assess the experience beyond the novelty experience, at scale and in-depth. At this point, the user researcher and data scientist were further embedded into the engineering team, close to product and design counterparts.

Mixing methods did prove to be a powerful practice to generate comprehensive and directional findings. The insights summaries containing both qualitative and quantitative findings, as well as a decision framework for A/B tests that was put in place during the feature iterations, were valued artifacts. As a result from the tight collaboration across disciplines, insights became a stronger partner in the product strategy work for collection and retrieval of content on the platform. We learned from our collaboration, for example the importance of keeping up the cadence of weekly syncs across product, design, engineering, data science, and user research disciplines. By embedding the data scientist closer to the engineers and the user researcher closer to product and design, taking part in regular discussions and joint settings of goals and guardrails, we found that trust and engagement was built (cf [16]). By this approach, we enabled a more "fearless organization" [6] for joint learning and innovation, where insight contributors had more agency in the planning and innovation, rather than finding data science and user research engaged in "policing" practices. We also found that the joint artifacts we made across data science and user research, such as planning docs and insights reports, supported the process even better than anticipated. This type of work, as well as continuously asking questions to each other about craft and implications, helped us cross the divide across competencies as we found that it's all too easy to assume knowledge across the qualitative and quantitative research disciplines.

4.3 Timing, Speed and Cost of Insights

We had set an ambitious insights plan with a strong focus on early insights. When comparing timelines to the previous re-design of the library feature we believe that it likely saved us time. As our process was iterative, with focus on early feedback and de-risking by multiple tests, this enabled us to detect risks and test crucial assumptions early on in the process when changes were easier to make. For example, by only building for one mobile operating system until the final tests were seen to be successful (instead of working across both core mobile operating systems at once), we were able to make changes with less time and cost.

4.3.1 Beta Testing. Our insights process sparked interest in the wider organization, and we were often engaged in discussions of when and how it makes sense to beta test novel features. In hindsight, we might have excluded the early beta test with the small set of users, as it had limited impact on the following build, although helpful to build trust and report in the core community. From the larger beta study, we did make important insights of the holistic experience over time, while still being able to change and adapt. Here, we would like to point out that such testing does not come without limitations. First of all, engineering and design time must be allocated to the processes, which is costly. Second, it must be an experience that users see a benefit or curiosity to opt into; it should be of significant experiential importance itself, or the data will likely be of limited use. A minor change to the user experience that the user can hardly detect may lead to limited behavioral changes and user reflections relevant to extract. We thus recommend only employing opt-in beta studies for clearly noticeable, substantial changes. Furthermore, one needs to be careful about conclusions by looking at opt-out numbers. Even if our rate of users staying in the beta was strikingly high, we should be careful about using this number as an argument for users being satisfied with the new Library. Opting out was likely harder to find than the opt-in message in Your Library. Another methodological aspect to point out (see for example [8]): because of network effects, public beta testers can increase users' valuation of a product and affect the outcomes.

Looking back, there would be further aspects that we would have changed in our beta approach, such as the data capture - for more directional and speedier impact, we could in hindsight have been more directional in our questions to users. Instead of a general rating, an absolute reference could have been used for more directionality, such as asking if the new experience was better than the previous Library experience or not, instead of a scale rating.

4.3.2 Tools. The redesign process generally adhered to the standard, long withstanding design process (see for example [9]) of an analysis phase (where we conducted for example the field study and data explorations), an ideation phase and an evaluation phase. However, a striking difference in current times are the tools available to speed up that process and deliver insights with more certainty. For example, using the LDA model for the beta test responses added speed. We later confirmed that the themes identified did indeed at large correlate with the identified themes from our semi-structured interviews. We look forward to what the future will bring in terms of even further speed and accuracy. Another aspect that improved insights quality and speed, was the easy development of interactive prototypes of mobile interaction concepts, which enabled more rapid and effective usability test iterations. The ability to also personalize these prototypes with a study participant's own Library content further enabled more directed and powerful qualitative user insights; it's one thing experiencing a hypothetical library, it's quite a different experience to interact with your own content in a new experience, especially considering the strong bond users have with their library content.

We recognize that all organizations do not have access to large data sets and the possibilities for experimentation that comes with that, but we encourage the use of mixed methods approaches with

the tools one might have available. We would like to stress the importance to iterate on concepts and break them down to smaller hypotheses which can be individually tested by also less data available; any customer-facing organization can learn plenty from talking to and observing users. Furthermore, beta practices where users can opt-in to a novel experience may still be available and can be less disruptive to a smaller business than A/B tests. From a qualitative research perspective, a user researcher in a small organization may easily feel overwhelmed by the needs for iterative usability testing - thus we also encourage user researchers to consider their research as a team sport and enable other disciplines to participate at all stages.

4.3.3 Timing. In addition to messaging the motivation and extent for the change to users, we want to stress that launch timing needs to be carefully chosen: too many changes at once across the platform leads to negative sentiment. Coordination across the organization is needed for introducing disruptive changes in user habits.

5 CONCLUSIONS AND NEXT STEPS

This case study presented examples of the qualitative and quantitative product insight work during the redesign of the Spotify's Your Library. We hope it has provided useful examples of delivering insights with depth and directionality throughout the redesign process of a feature associated with strong user habits.

We built the Library with the intention to be able to adapt and evolve for future growth. One such growth opportunity was to add new content types to the platform, which recently was made when Spotify added Audiobooks to its content repertoire [21] including the possibility to save books to users' personal library. With this, and other developments, we are still learning how listeners are using and experiencing the new Library and how we can best approach the experience across multiple content types, and multiple segments of users. We will continue to work in close collaboration across disciplines to bring further improvements to the experience.

ACKNOWLEDGMENTS

We want to acknowledge the work by the collaborating insights, product, design and tech teams.

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Received 13 October 2022; accepted 30 November 2022